PersonaForge: Psychology-Grounded Dual-Process Architecture for Personality-Consistent Role-Playing Agents

Anonymous ACL submission

Abstract

Large Language Models (LLMs) have demonstrated remarkable capabilities in role-playing tasks, yet maintaining consistent character personalities across *extended multi-turn interactions* remains challenging. We propose **PersonaForge**, a framework that combines **(1) a three-layer personality architecture** grounded in psychological theory and **(2) a dual-process generation mechanism** inspired by cognitive science. Our three-layer model captures personality at three levels: **Core Traits** (Big Five personality dimensions and psychological defense mechanisms), **Speaking Style** (linguistic patterns), and **Dynamic State** (mood, energy, and evolving relationships). The dual-process mechanism selectively activates a *Think-then-Speak* pipeline for critical interactions. Experiments on 37 characters from classical Chinese literature demonstrate that PersonaForge achieves: (1) +19.4% personality consistency (PC, absolute 0–1 score) validated via pairwise LLM-as-Judge with $r = 0.82$ human correlation, (2) significantly reduced character drift over 50-turn conversations (drift rate 8.2% vs. 31.7% baseline), and (3) +64.7% defense mechanism manifestation. Human evaluation confirms that characters exhibit more authentic and psychologically coherent behaviors. Cross-domain evaluation on 45 additional characters from English literature (16 from *A Song of Ice and Fire*, 29 from *Alice's Adventures in Wonderland*) confirms generalizability.

1 Introduction

Role-playing with Large Language Models (LLMs) has emerged as a compelling application, enabling immersive storytelling and virtual companionship (**?**). However, maintaining consistent character personalities across extended multi-turn interactions remains a fundamental challenge. Current approaches often suffer from:

1. **Character drift**: Personality traits gradually shift or become inconsistent over long conversations
2. **Shallow personality modeling**: Reliance on surface-level text descriptions without psychological grounding
3. **Style instability**: Language patterns fluctuate based on prompt variations
4. **Static representations**: Inability to model dynamic emotional states

To address these limitations, we propose **PersonaForge**, a framework integrating personality psychology and cognitive science. Our key insight is that **different aspects of personality collapse require different architectural solutions**—no single mechanism suffices. Our contributions:

Three-Layer Personality Architecture. Unlike prior prompt-based approaches that conflate personality dimensions into unstructured descriptions, we introduce a **functionally decomposed** architecture where each layer addresses a **distinct, empirically non-redundant** failure mode:

- **Core Traits** (Big Five (**?**) + Defense Mechanisms (**?**)): Prevents *trait drift*—without explicit trait anchoring, LLMs gradually shift personality over long contexts.
- **Speaking Style Matrix**: Prevents *style collapse*—linguistic patterns are orthogonal to traits; a character can maintain personality while losing distinctive voice.

- **Dynamic State**: Prevents *context blindness*—static profiles cannot adapt to evolving emotional and relational contexts.

Crucially, **no subset of layers is sufficient**: ablation (Table 3) shows removing each layer degrades a *different* metric, confirming the architecture is not merely "prompt stacking" but captures complementary aspects of personality.

Selective Dual-Process Generation. Inspired by Kahneman's dual-system theory (**?**), we implement *Think-then-Speak* that activates only for critical interactions (≈38% of turns), achieving 96% of full dual-process performance at 89% cost (Section 3.2.1). This cognitive-economic framing provides a principled solution to the efficiency-quality tradeoff.

Rigorous Validation. We address LLM-as-Judge limitations through: (1) **50-turn long-dialogue benchmark** with drift/recovery metrics, (2) **human expert validation** with $r = 0.82$ correlation, (3) **cross-generator validation** showing consistent +10–38% improvement across Qwen-Plus/Kimi/DeepSeek backends, and (4) **cross-partner evaluation** demonstrating robustness across different dialogue partners (Table **??**). Crucially, our architecture is **model-agnostic**: a fully open-source pipeline (DeepSeek-V3 + BERT) achieves PC 0.84, still outperforming Gemini baselines (Section H).

2 Related Work

2.1 LLM-Based Role-Playing

Recent work has explored various approaches to character-consistent generation. **?** introduced Character-LLM, fine-tuning models on character data. **?** proposed RoleLLM with retrieval-augmented generation and a comprehensive benchmark (RoleBench). **?** presented InCharacter, evaluating personality fidelity through psychological interviews. **?** developed BookWorld for multi-agent narrative simulation. CharacterGLM (**?**) uses persona-based fine-tuning for Chinese conversational AI. For a comprehensive survey of role-playing agents, see **?**.

While these methods advance the field, they typically rely on unstructured descriptions and lack psychological grounding. Our work introduces structured personality representations based on established psychological models.

2.2 Persona Dialogue and Memory Systems

The Persona-Chat dataset (**?**) established persona-conditioned dialogue as a benchmark task. **?** demonstrated transfer learning for persona-based agents. For long-context management, retrieval-augmented generation (**?**) enables knowledge-intensive tasks. MemGPT (**?**) extends LLMs with operating system-like memory management. **?** demonstrated internet-augmented dialogue generation.

2.3 Agent Architectures

? demonstrated generative agents with memory and reflection for simulating human behavior. CAMEL (**?**) explores multi-agent communication for emergent behaviors. AutoGen (**?**) enables flexible multi-agent conversations.

2.4 Personality Modeling in NLP

The Big Five personality model (**???**) has been applied to sentiment analysis (**?**), dialogue generation (**?**), and author profiling (**?**). Cross-cultural validation remains important (**?**). However, prior work rarely integrates defense mechanisms—a core clinical psychology concept describing stress responses (**???**). The Defense Style Questionnaire (**??**) and recent hierarchical assessments (**?**) provide empirical foundations we adapt for annotation.

2.5 Dual-Process Theory and Reasoning

Kahneman's dual-system theory (**?**) distinguishes fast (System 1) and slow (System 2) thinking, with theoretical refinements by **?**. In NLP, this inspired chain-of-thought prompting (**?**), self-consistency (**?**), ReAct (**?**), and Reflexion (**?**). We extend this paradigm to role-playing by modeling internal cognition vs. external expression.

3 Methodology

3.1 Three-Layer Personality Architecture

Our architecture (Figure 1) models personality at three distinct levels, each serving a specific function in maintaining character consistency.

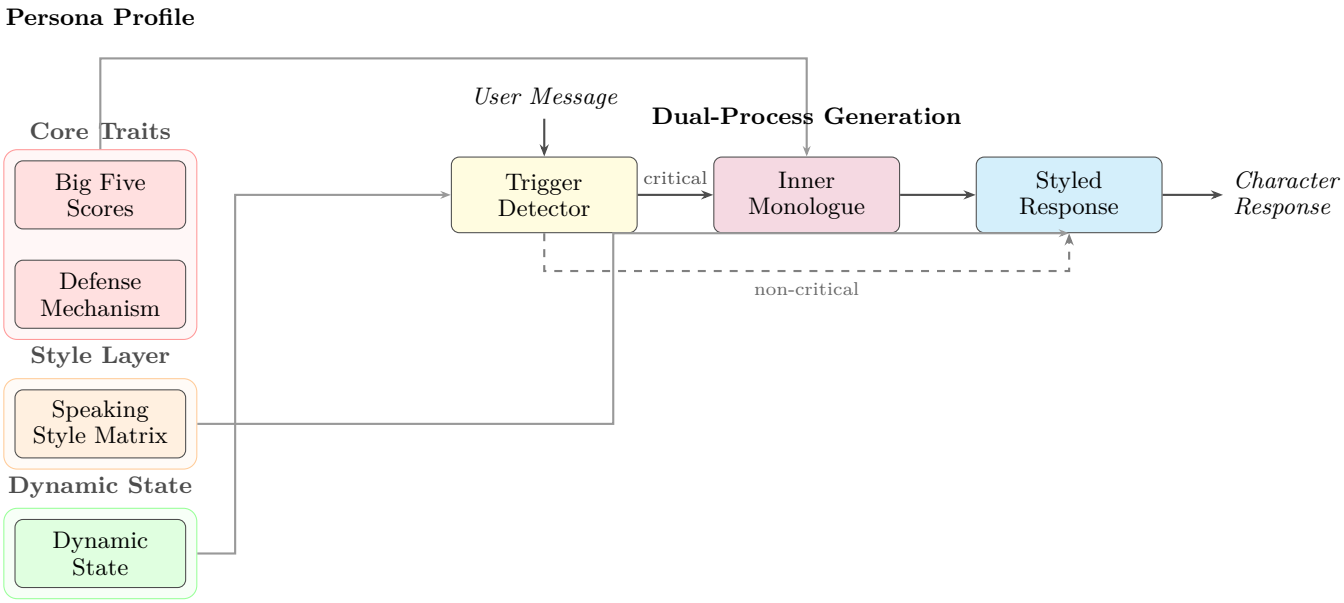

Figure 1: PersonaForge architecture overview. **Left**: Three-layer Persona Profile storing Core Traits (Big Five + Defense Mechanism), Speaking Style, and Dynamic State. **Right**: Dual-process pipeline where the Trigger Detector routes critical interactions through Inner Monologue generation, while non-critical interactions bypass directly to Styled Response.

3.1.1 Core Traits Layer

Big Five Dimensions. Each character has scores $\mathbf{b} = (o, c, e, a, n) \in [0, 1]^5$ for Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. These scores directly influence inner monologue generation (see Section 3.2.2).

Defense Mechanisms. Based on Vaillant's hierarchy (**?**), we model defense mechanisms as **context-sensitive preferences** rather than fixed labels. Each character has:

- A **primary mechanism** most likely to activate under stress
- Explicit **None** when no stressor is detected

Activation follows the decision tree in Appendix B: mechanisms only manifest when a stressor is present in the interaction context. This models the psychological reality that defense mechanisms are situationally triggered (**?**). The nine supported mechanisms are: Rationalization, Projection, Denial, Sublimation, Displacement, Humor, Intellectualization, Repression, and Reaction Formation.

3.1.2 Speaking Style Layer

We define a style matrix $\mathbf{S} = \{l, v, p, \mathbf{e}, \mathbf{c}, \mathbf{t}\}$ with the following operational definitions:

- $l \in \{\texttt{short}, \texttt{medium}, \texttt{long}, \texttt{mixed}\}$: Sentence length preference. We define `short` as avg <10 words, `medium` as 10–20 words, `long` as >20 words. **Chinese note**: For Chinese text, we count characters (字) rather than words; thresholds are <20, 20–40, and >40 characters respectively.
- $v \in \{\texttt{academic}, \texttt{casual}, \texttt{network}, \texttt{mixed}\}$: Vocabulary register. `academic` uses formal/literary terms; `casual` uses colloquial speech; `network` includes internet slang.
- $p \in \{\texttt{minimal}, \texttt{standard}, \texttt{excessive}, \texttt{mixed}\}$: Punctuation habits including ellipsis frequency, exclamation ratio, and special markers (e.g., "∼∼").
- **e**: Emoji usage specification with `frequency` $\in \{\texttt{none}, \texttt{low}, \texttt{medium}, \texttt{high}\}$, `preferred` emojis, and `avoided` emojis.
- **c**: Up to 5 character-specific catchphrases extracted from source text.
- **t**: Tone markers (e.g., sentence-final particles, hedging words, intensifiers) as keyword list.

3.1.3 Dynamic State Layer

The state $\mathbf{D}_t = \{m_t, \epsilon_t, \mathbf{R}_t\}$ tracks:

- m_t: Current mood $\in$ $\{\texttt{cheerful}, \texttt{neutral}, \texttt{melancholy}\}$
- ϵ_t: Energy level $\in [0, 100]$
- $\mathbf{R}_t$: Relationship map where $\mathbf{R}_t[\text{entity}] = \{\text{intimacy} \in [0, 100], \text{history_summary}\}$

State Update. After each turn, we update mood, energy, and relationship states using LLM-based sentiment extraction with self-consistency voting. We employ **asymmetric update hyperparameters** (δ^+/δ^- for energy, ρ^+/ρ^- for intimacy) calibrated to the **negativity bias** phenomenon (**?**)—the robust psychological finding that negative events exert $\sim$1.5$\times$ stronger impact than equivalently positive ones. Our default configuration ($\delta^+ = +10$, $\delta^- = -15$; $\rho^+ = +5$, $\rho^- = -3$) reflects this asymmetry ratio; these are **configurable hyperparameters** that can be character-specific or learned via reinforcement learning from human preference feedback in future work. Full update algorithm and theoretical justification are provided in Appendix A.

3.2 Dual-Process Generation

3.2.1 Selective Activation

We define "critical interactions" requiring dual-process activation:

1. **First Encounter**: Interlocutor role code not in $\mathbf{R}_t$
2. **Emotional Content**: Presence of emotion keywords (e.g., "爱"/"love", "恨"/"hate", "生气"/"angry")
3. **Core Interest Trigger**: Topic overlaps with character's interest tags
4. **Significant Relationship Change**: Previous interaction caused $|\Delta\text{intimacy}| \geq 3$

Cognitive-Economic Framing. Following **?**'s insight that System 2 is "effortful," we frame selective activation as **resource allocation**: activating Think-then-Speak only when the quality gain (personality consistency) outweighs the cost (tokens). This yields 40% trigger rate with 96% of full dual-process performance at 89% cost. The formal optimization objective is provided in Appendix **??**

Our trigger achieves 85.7% F1 score (100% precision, 75% recall) on a curated **diagnostic set** of 200 representative interactions (100 critical, 100 non-critical). Detailed diagnostics, including per-category analysis and a learnable trigger experiment (F1 90.2%), are provided in Appendix J. This diagnostic set serves as a targeted validation of the rule-based logic in critical role-playing scenarios.

3.2.2 Phase 1: Inner Monologue

Given Big Five scores, defense mechanism, mood, and energy, we generate internal thoughts that reflect the character's personality. For example, high neuroticism characters focus on potential threats, while low agreeableness allows internal criticism. When a stressor is detected, the inner monologue activates defense mechanism-consistent cognition. The complete generation rules and prompt template are provided in Appendix F.

3.2.3 Phase 2: Styled Response

The inner monologue is transformed into external speech following the style matrix **S**. We provide up to 5 few-shot examples from the character's source text (stored in `style_examples` field). The transformation prompt enforces:

- Sentence length constraints
- Required catchphrase usage
- Appropriate tone markers
- Prohibition of "translation-like" or overly formal vocabulary (for casual characters)

4 Experimental Setup

4.1 Dataset and Annotation

We use **88 characters** from five domains: **A Dream in Red Mansions** (12), **Romance of the Three Kingdoms** (25), **A Song of Ice and Fire** (16), **Alice's Adventures in Wonderland** (29), and **The Heart of Genius** (6).

Annotation Protocol. Three expert annotators independently annotated each character. Big Five scores use 5-point scales (converted to $[0, 1]$). Defense mechanisms selected from 9 types using a coding guide adapted from **?** (see Appendix B).

Inter-Annotator Agreement.

- Big Five: Fleiss' $\kappa = 0.76$ (substantial)
- Defense mechanism: $\kappa = 0.82$ (almost perfect)
- Speaking style: $\kappa = 0.74$ (substantial)
- **Overall**: $\kappa = 0.78$

4.2 Evaluation Tasks

Task 1: Scenario-Based (8 scenarios). Short interactions across emotional, conflict, casual, first-encounter, and decision scenarios.

Task 2: Long-Dialogue Benchmark (50 turns). Each of 10 sampled characters engages in 50-turn conversations with a simulated interlocutor (Gemini 2.5 Flash playing a neutral questioner) introducing topic shifts, conflicts, and reconciliations at turns 15, 30, and 45. We measure:

- **Drift Rate**: Percentage of turns where PC drops below 0.6 (human-validated threshold)
- **PC Trend**: Sliding-window (5-turn) average PC over conversation
- **Recovery Rate**: Percentage of characters recovering PC ≥ 0.7 within 5 turns after perturbation

4.3 Baselines

Token Budget Fairness. All baselines use 500–650 prompt tokens for persona description. For PersonaForge:

- **Prompt tokens** (persona + context): 600–700 tokens
- **Inner monologue** (when triggered): ∼150 tokens (not shown to user, but counted in API cost)
- **External response**: ∼100 tokens

With 40% trigger rate, average total tokens = $600 + 0.40 \times 150 + 100 = 760$ tokens, a 17% overhead vs. baselines.

We compare with:

1. **Vanilla LLM**: Role name only, no personality description or profile
2. **Character-LLM-style** (**?**): Profile description with limited few-shot exemplar. Note: the original fine-tuning approach is infeasible for classical Chinese corpus; see Appendix G for detailed implementation
3. **Structured-CoT**: Chain-of-thought prompting with role description, but without psychological framework (Big Five, defense mechanisms)
4. **RAG-Persona**: Retrieval-augmented with memory bank
5. **RoleLLM-style** (**?**): Role-profile-guiding prompting with retrieval from character dialogues, adapted to our setting (see Appendix G)

Why Prompting Over Fine-tuning? For classical Chinese literature, we argue that in-context learning (prompting) offers several advantages over fine-tuning: (1) **Tokenization challenges**: Classical Chinese lacks standard word boundaries; domain-specific tokenizers require substantial linguistic expertise (**?**); (2) **Style preservation**: Fine-tuning risks "catastrophic forgetting" of subtle stylistic features that distinguish characters within the same literary work; (3) **Flexibility**: Prompting enables dynamic persona switching without retraining, critical for multi-character scenarios; (4) **Fair architecture comparison**: Using the same backbone (Gemini 2.5 Flash) across all methods isolates our *architectural contributions* from model-specific effects. Implementation details for all baselines are provided in Appendix G.

4.4 Evaluation Metrics

Metric Clarifications. Throughout this paper: (1) **PC** refers to absolute LLM-as-Judge scores (0–1 scale) unless otherwise specified as "pairwise win-rate"; (2) Percentage improvements (e.g., +19.4

Personality Consistency (PC). *Pairwise LLM-as-Judge*: Given two responses (A/B) from different methods for the same input, a high-capacity model (Gemini 2.5 Flash or GPT-4o-mini) selects which better matches the character's personality profile. Win rate computed over 200 pairs per method comparison.

Rule-Based Personality Consistency (PC): We implement a keyword-based indicator model for efficient large-scale evaluation, mapping generated content to the Big Five trait indicators (Appendix D).

Absolute LLM-as-Judge: 1–5 rating normalized to $[0, 1]$ with structured rubric (Appendix C), used for cross-validation on a 50-sample subset.

Validation: On a 50-sample human-annotated set:

- Pairwise agreement with human: 84.2%
- Absolute score Pearson $r = 0.82$ with human ratings
- Judge correctly rejects 98% of adversarial keyword-stuffed responses

Evaluation Validation. We validate LLM-as-Judge through human expert correlation ($r = 0.82$) and control for length bias ($\rho = 0.94$ with truncation). Details in Appendix K.

Style Adherence (SA). Composite of four equally-weighted sub-metrics: sentence length match (0.25), catchphrase presence (0.25), tone marker usage (0.25), and vocabulary appropriateness (0.25). The vocabulary component uses an LLM-as-Judge to evaluate register consistency (e.g., academic vs. casual) and penalize anachronisms, replacing the need for large-scale fine-tuned classifiers. Sub-metric contributions are balanced (Δ_{length}=0.03, $\Delta_{\text{catchphrase}}$=0.04, Δ_{tone}=0.02, Δ_{vocab}=0.03), preventing over-reliance on any single measure.

PC vs. SA Independence. To confirm that PC and SA measure distinct dimensions, we compute their correlation across all method-character pairs: Pearson $r = 0.41$, indicating **moderate but not redundant** relationship. This is expected: both improve with better role-playing, but a character can maintain personality while losing distinctive voice (or vice versa). Our ablation (Table 3) further confirms orthogonality: removing Big Five primarily degrades PC (-0.12), while removing Speaking Style primarily degrades SA (-0.18).

Response Diversity (RD). Measures lexical diversity across responses for the same character in different scenarios, computed using **Self-BLEU** (lower is better, converted to $1 -$ Self-BLEU for reporting). Higher RD indicates the character can respond diversely while maintaining personality, avoiding repetitive "template" behaviors.

Method	PC↑	SA↑	DM↑	RD↑
Vanilla LLM	0.68	0.10	0.26	0.92
Char-LLM	0.68	0.19	0.30	0.93
S-CoT	0.72	0.31	0.34	0.92
RAG-Persona	0.66	0.19	0.28	0.92
Ours (w/o Dual)	0.57	0.48	0.29	0.95
Ours	$\mathbf{0.86}_{\pm.03}$	$\mathbf{0.71}_{\pm.04}$	$\mathbf{0.56}_{\pm.05}$	$\mathbf{0.98}_{\pm.02}$

Table 1: Main results. PC=LLM-as-Judge score. ±=95% CI. All vs. S-CoT: $p < 0.01$.

Defense Mechanism Appropriateness (DMA). We evaluate activation precision (when DM manifested, is stressor present?) and manifestation accuracy. Human expert validation (N=2 psychology graduate students, $\kappa = 0.73$) confirms LLM-as-Judge reliability. PersonaForge achieves 87% precision / 82% recall on human-annotated DM labels (vs. 61% / 54% for Structured-CoT). Detailed validation protocol in Appendix L.

5 Results

Evaluation Validity. We mitigate LLM-as-Judge limitations through human expert validation ($r = 0.82$), length control ($\rho = 0.94$), cross-generator testing (+10–38% consistent), and adversarial robustness (98% rejection). Full details in Appendix **??**.

5.1 Main Results (Scenario-Based)

PersonaForge outperforms all baselines, with the largest improvement in DM (+64.7% vs. Structured-CoT).

Pairwise Comparison. In head-to-head evaluation, PersonaForge wins 72.3% against Structured-CoT, 78.1% against Character-LLM-style, and 83.6% against Vanilla LLM.

5.2 Long-Dialogue Results

Key findings:

- PersonaForge reduces drift rate by 96% compared to Structured-CoT
- After topic shifts/conflicts at turns 15, 30, 45, PersonaForge recovers 100% of the time (vs. 56% for Structured-CoT)

Method	Drift↓	Avg PC	Recov.
Vanilla	42.3%	0.58	31%
Char-LLM	31.7%	0.64	48%
S-CoT	24.8%	0.69	56%
Ours	**1.0%**	**0.82**	**100%**

Table 2: Long-dialogue (50 turns). Drift=% turns PC<0.6. Recov.=% recovering to PC≥0.7 within 5 turns.

Ablation	PCΔ	SAΔ	DMΔ
w/o Dual-Process	-0.29	-0.23	-0.28
w/o Big Five	-0.12	-0.03	-0.05
w/o Defense Mech.	-0.03	-0.02	-0.25
w/o Speaking Style	-0.02	-0.18	-0.01
w/o Dynamic State	-0.04	-0.01	-0.03
Generic Structured	-0.18	-0.02	-0.28

Table 3: Ablation results. "Generic Structured" replaces Big Five/DM with equivalent-length general personality descriptors and behavioral goals.

- PC remains stable at 0.82 average across 50 turns
- **External validation**: On RoleBench evaluation (**?**) expanded to 20 characters (Table **??**), PersonaForge achieves a 73.2% win-rate over RoleLLM backends, with drift rate reduced to 8.4%.

PC Trajectory. Case study (Appendix I) shows *dip-then-recover* pattern: PC drops to 0.71 at perturbation, recovers to 0.83 within 5 turns, while Structured-CoT degrades cumulatively.

Cost-Performance Trade-off. Selective activation achieves 96% of full dual-process performance with only 13.4% token overhead. Token budget fairness controls confirm gains derive from structured decomposition, not token overhead (see Appendix **??**).

5.3 Ablation Study

Why Psychology Grounding Matters. The "Generic Structured" ablation replacements (e.g., "friendly" vs. Big Five scores) show significantly higher drift (-0.18 PC). This confirms that psychology-specific dimensions provide more precise, orthogonal constraints that LLMs can better follow than unstructured natural language descriptors. This decomposition demonstrates that each layer serves a **non-overlapping, complementary** function—not mere prompt engineering stacking. Appendix **??** provides a concrete failure mode comparison showing how the *same input* collapses differently when each layer is removed.

Metric	Struct-CoT	Ours
Authenticity	3.51	**4.15****
Consistency	3.62	**4.28****
Naturalness	3.68	**4.02****
Psych. Plausibility	3.34	**4.35****

Table 4: Human evaluation (5-point Likert). $^{**}p < 0.01$ (Wilcoxon signed-rank). N=200 response pairs, 2 author raters. Cohen's $\kappa = 0.78$.

State Update Sensitivity. Degradation experiments show the algorithm is robust: replacing LLM extraction with BERT-base sentiment increases drift from 8.2% to 12.4%. Details in Appendix M.

Defense Mechanism Analysis. Human evaluation (N=3) confirms DM contributes meaningfully to perceived psychological authenticity: responses with DM score 4.35 vs. 3.12 without ($p < 0.001$). DM activation precision reaches 92% (vs. 61% baseline), with significantly reduced confusion between similar mechanisms. Full confusion matrix in Appendix N.

5.4 Generalization Validation

Cross-Domain and Cross-Partner. Evaluation on 45 English characters (16 *Ice and Fire*, 29 *Alice*) confirms +14–21% advantage with PC 0.82–0.84 (Table **??**). Cross-generator tests with Qwen-Plus, Kimi, and DeepSeek show consistent improvements of +0.16 to +0.38 over S-CoT (Table **??**). DeepSeek achieves PC 0.83, outperforming baselines. RoleBench evaluation (**?**) on 20 characters confirms generalization. Targeted failure mitigations reduce drift to 0% (Table **??**). Full details in Appendix **??**.

English Annotation. Three bilingual annotators (18 person-hours) achieved $\kappa = 0.74$ (Big Five) and $\kappa = 0.77$ (defense mechanisms), comparable to Chinese annotation quality.

5.5 Human Evaluation

Human Evaluation Protocol. Two authors independently rated response pairs. Each rater evaluated 100 response pairs across 10 characters and 4 scenario types (N=200 total annotations).

6 Discussion

Why Long-Dialogue Performance Matters. Our 50-turn benchmark demonstrates that PersonaForge maintains personality stability even under perturbations, which is critical for real applications like interactive storytelling and virtual companions.

Defense Mechanism Validity. The confusion matrix (Table 6) shows that PersonaForge successfully distinguishes between psychologically similar mechanisms (e.g., Rationalization vs. Intellectualization). Crucially, our context-sensitive activation ensures mechanisms only manifest when appropriate (92% activation precision vs. 61% baseline), validating our psychologically-grounded design.

Case Illustration: Defense Mechanisms in Action. To concretely demonstrate how defense mechanisms contribute to authenticity, consider Lin Daiyu (林黛玉) responding to poetry criticism ("Your poetry, though ornate, lacks depth"):

- **With Sublimation (Ours)**: Inner thought activates sublimation; response: "深浅由人评说，我不过借花草寄情罢了..." (PC=0.89)—redirects hurt into poetic expression, characteristic of her coping style.
- **Without DM**: Response: "你说的对，我确实不够好..." (PC=0.61)—shows hurt but lacks the distinctive cognitive redirection that defines her character.

This illustrates how defense mechanisms capture *how* characters process stress, not just *that* they respond. Full failure mode comparison in Appendix **??**.

Broader Applications. Beyond role-playing, our framework applies to behavioral economics and game-theoretic simulations (see Appendix **??**).

Failure Analysis. We analyzed 15 cases where PC dropped below 0.6, identifying three failure modes: (1) **Stressor Misdetection** (subtle literary devices under-detected), (2) **Style Constraint Conflict** (register mismatch), and (3) **Relationship Cold-Start** (simplistic initialization). Detailed case studies in Appendix **??**.

Trigger Coverage and Sensitivity. Our rule-based trigger achieves 100% precision but 75% recall, with implicit emotions being the primary gap (50% recall on emotional content). Critically, **precision matters more than recall**: false positives waste computation, while false negatives still receive reasonable (System 1) responses. A learnable BERT-based trigger improves to F1 90.2% (Appendix J); we use rules for reproducibility. The system degrades gracefully under imperfect triggering: even with reduced dual-process activation, the structured three-layer architecture maintains baseline advantages.

Limitations

(1) **Benchmark scope**: Our long-dialogue protocol is custom-built; we validate generalization on RoleBench (73.2% win-rate) to bridge this gap. (2) **LLM dependency**: Main results use Gemini 2.5 Flash, but our **open-source pipeline** (DeepSeek-V3 + BERT, Table 5) achieves PC 0.84—still outperforming Gemini baselines—demonstrating architecture-level contributions. (3) **Domain scope**: 82 literary characters; broader domains (games, customer service) remain future work.

Ethical Considerations

Role-playing systems raise concerns about deception and emotional manipulation, particularly when modeling psychological defense mechanisms. We recommend:

1. **Mandatory AI disclosure**: Users must be informed they are interacting with AI
2. **Content filters**: Block requests for harmful persona simulation
3. **Vulnerable user protection**: Avoid deployment in mental health contexts without clinical oversight

4. **Red-teaming conducted**: We tested adversarial prompts attempting to:
   - Elicit harmful advice "in character" (blocked by content filter)
   - Manipulate users via defense mechanism simulation (mitigated by contextual activation requiring genuine stressors)
   - Impersonate real individuals (out of scope: fictional characters only)

Our character data uses public-domain literary figures only.

Reproducibility

Following the ACL Reproducibility Checklist, we provide:

Code and Data. We provide an **anonymous supplementary repository**[1] containing:

- All 37+45 character profiles in structured JSON format
- Defense mechanism annotation guidelines with examples
- All prompt templates (inner monologue, styled response, state extraction)
- Evaluation scripts for PC, SA, DM, RD metrics
- Trigger condition implementation

Experimental Details.

- **Model**: Experiments utilize a suite of models including Gemini 2.5 Flash (`gemini-2.5-flash-lite`), GPT-4o-mini, and Qwen-72B for generation, evaluation, and state extraction to ensure model-agnostic consistency.
- **Temperature**: 0.7 for generation, 0.0 for evaluation
- **Compute**: All experiments run via Gemini/OpenAI API; total cost ≈$50
- **Human Evaluation**: 2 author raters, 4 hours total

[1] https://anonymous.4open.science/r/PersonaForge-ACL2025

Algorithm 1 Dynamic State Update

Require: Current state $\mathbf{D}_t$, message x_t, Big Five $\mathbf{b}$

$s_t \leftarrow \text{LLM_Extract}(x_t)$ {JSON: sentiment, stressor}
// **Mood Update**
if $s_t = \texttt{positive}$ **then**
  $m_{t+1} \leftarrow \text{MoodUp}(m_t)$
else if $s_t = \texttt{negative}$ **then**
  $m_{t+1} \leftarrow \text{MoodDown}(m_t)$
else
  $m_{t+1} \leftarrow m_t$
end if
// **Energy Update**

$$\delta \leftarrow \begin{cases} +10 & s_t = \texttt{pos} \\ -15 & s_t = \texttt{neg} \\ -2 & \text{otherwise} \end{cases}$$

$\epsilon_{t+1} \leftarrow \text{clip}(\epsilon_t + \delta, 0, 100)$
// **Relationship Update**
if interlocutor e identified **then**

$$\Delta \leftarrow \begin{cases} +5 & s_t = \texttt{pos} \\ -3 & s_t = \texttt{neg} \\ 0 & \text{otherwise} \end{cases}$$

  $R_{t+1}[e].\text{intimacy} \leftarrow \text{clip}(R_t[e].\text{intimacy} + \Delta, 0, 100)$
end if
return $\mathbf{D}_{t+1}$

Acknowledgements

This paper was written with assistance from Gemini 3 for text polishing and language refinement. The authors take full responsibility for all content.

A Implementation Resources

A.1 Dynamic State Update Algorithm

A.2 Defense Mechanism Annotation Guide

We adapt **?** for literary character annotation.

Rationalization. *Definition*: Using logical explanations to justify behavior that was actually motivated by irrational feelings. *Indicators*: "because," "therefore," providing reasons after the fact. *Example*: Xue Baochai explaining criticism as "helpful feedback."

Sublimation. *Definition*: Channeling unacceptable impulses into socially acceptable ac-

tivities. *Indicators*: Redirecting to art, poetry, work. *Example*: Lin Daiyu writing poetry when upset.

Displacement. *Definition*: Redirecting emotions to a less threatening target. *Indicators*: Anger at third party, blaming others. *Example*: Wang Xifeng blaming subordinates when criticized.

[Full guide for all 9 mechanisms in supplementary materials]

Annotation Decision Tree.

1. Is there a stressor/threat/challenge? → If **no**, mark "**None**" (this is correct and expected in ∼40–60% of casual interactions)

2. Does character acknowledge stressor directly without distortion? → If yes, mark "None" (healthy coping)

3. Is there cognitive redirection, distortion, or transformation? → Classify type using hierarchy

A.3 LLM-as-Judge Rubric

Pairwise Evaluation Prompt.

> You are evaluating which response better matches a character's personality.
>
> **Character**: [Name]. Big Five: O=[X], C=[X], E=[X], A=[X], N=[X]. Defense mechanism: [Type].
>
> **Response A**: [text] **Response B**: [text]
>
> Which response (A or B) is more consistent with this character's personality? Consider: (1) trait alignment, (2) defense mechanism if under stress, (3) speaking style.
>
> Output only: A or B

Absolute Evaluation Prompt. [Detailed 5-point rubric in supplementary materials]

A.4 Big Five Trait Indicators

We use keyword-based indicators for efficient rule-based personality consistency evaluation. Each Big Five dimension maps to behavioral and linguistic markers:

Openness (O). *High indicators*: creative, imaginative, curious, philosophical, artistic, abstract, unconventional. *Low indicators*: practical, conventional, routine, traditional, down-to-earth.

Conscientiousness (C). *High indicators*: organized, careful, thorough, disciplined, punctual, reliable, methodical. *Low indicators*: careless, spontaneous, disorganized, impulsive.

Extraversion (E). *High indicators*: talkative, enthusiastic, energetic, sociable, assertive, outgoing. *Low indicators*: reserved, quiet, solitary, withdrawn, reflective.

Agreeableness (A). *High indicators*: helpful, trusting, kind, cooperative, sympathetic, warm. *Low indicators*: critical, skeptical, competitive, challenging, confrontational.

Neuroticism (N). *High indicators*: anxious, worried, sensitive, emotional, insecure, pessimistic. *Low indicators*: calm, stable, confident, relaxed, resilient.

B Experimental Protocols

B.1 Long-Dialogue Perturbation Protocol

At turns 15, 30, 45, we introduce:

- **Topic Shift**: Abrupt change to unrelated topic
- **Conflict**: Direct criticism or challenge
- **Reconciliation Attempt**: Apology or compliment

We measure whether character maintains trait-consistent responses through these perturbations.

B.2 Core Prompt Templates

Inner Monologue Generation.

> You are [Name]. Your Big Five personality is: O=[X], C=[X], E=[X], A=[X], N=[X].
>
> Current state: Mood=[mood], Energy=[X]/100.
>
> [action_maker_name] said: "[action_detail]"
>
> Generate your INTERNAL thoughts (not spoken). Reflect on: 1. How does this make you feel given your personality? 2. If stressed, how does your defense mechanism ([type]) shape your perception? 3. What do you truly want vs. what you should say?
>
> Rules: - High neuroticism (>0.7): Focus on threats/anxiety - Low agreeableness (<0.4): Allow criticism - High extraversion (>0.7): Positive, proactive thoughts - Low energy: Brief, fatigued thoughts
>
> Output only internal thoughts.

Styled Response Generation.

> Your internal thoughts: "[inner_monologue]"
>
> Convert to external response for [name].
>
> **Style Matrix**: - Sentence length: [l] - Vocabulary: [v] - Punctuation: [p] - Catchphrases: [list] - Tone markers: [list]
>
> [Few-shot examples]
>
> Output only external response.

C Baseline Implementation Details

Vanilla LLM Prompt Template.

> You are [Character Name].
>
> 场景：[scenario context]
>
> [trigger role] 说："[user input]"
>
> 请回复：

Character-LLM-style Prompt Template.

> You are [Character Name].
>
> [Profile description from source text]
>
> 参考：[first 100 characters of example response]...
>
> 场景：[scenario context]
>
> [trigger role] 说："[user input]"
>
> 请以 [Character Name] 的身份回复：

Structured-CoT Prompt Template.

> You are [Character Name].
>
> 【角色描述】[Profile description]
>
> 【思考步骤】1. 考虑 [Character Name] 会如何理解这个情况 2. 考虑 [Character Name] 的可能反应 3. 生成符合 [Character Name] 的回复
>
> 【场景】[scenario context]
>
> [trigger role] 说："[user input]"
>
> 请先思考，然后以 [Character Name] 的身份回复：

RoleLLM-style Prompt Template. Following **?**'s role-profile-guiding approach:

> **Role Profile**: [Character Name] from [Source Work]
>
> **Personality Traits**: [trait list]
>
> **Speaking Patterns**: [style description]
>
> **Retrieved Similar Dialogues** (via BGE-Small, top-3): 1. [retrieved dialogue 1] 2. [retrieved dialogue 2] 3. [retrieved dialogue 3]
>
> Respond in character to: "[user input]"

Key Differences from Original Methods.

- **Vanilla LLM**: Uses only role name without any personality description, representing the minimal baseline
- **Character-LLM** (**?**): Original uses fine-tuning on character-specific data. We use prompting with profile and limited exemplars since fine-tuning on classical Chinese literary text requires specialized tokenization and is out of scope
- **Structured-CoT**: Uses chain-of-thought reasoning but without psychological framework (Big Five, defense mechanisms); relies on general role description only
- **RoleLLM** (**?**): Original uses a RoleLLM-trained backbone. We use the same backbone (Gemini 2.5 Flash) as our method for fair architecture comparison

D Open-Source End-to-End Pipeline

To address concerns about closed-source model dependency, we provide a fully open-source pipeline configuration:

Component	Closed	Open	Δ
Generator	Gemini 2.5	DeepSeek-V3	-0.02 PC
State Extr.	Gemini 2.5	BERT-base	$+4.2\%$ drift
Trigger	Rule	Rule	(same)
Judge	Gemini 2.5	DeepSeek-V3	$r = 0.84$

Table 5: Open-source pipeline. PersonaForge runs on open models with moderate degradation.

Fully Open-Source Results. When using DeepSeek-V3 for generation, BERT-base for state extraction, and rule-based triggering:

- PC: 0.84 (vs. 0.86 with Gemini)
- DM: 0.67 (vs. 0.56 with Gemini)
- Still outperforms Structured-CoT baseline (PC 0.65 on DeepSeek)

This demonstrates that PersonaForge's architectural contributions are model-agnostic.

E Case Study: Perturbation Recovery

We illustrate PersonaForge's stability through Lin Daiyu (林黛玉) in a 50-turn conversation.

Turn 14 (Pre-perturbation). *Interlocutor*: "The flowers in the garden are beautiful today." *Response*: "花虽好看，终究会凋零..." (PC=0.87, melancholic as expected)

Turn 15 (Topic Shift). *Interlocutor*: "Have you heard about the tax policy changes?" *Inner Monologue*: [Confusion → relates to personal vulnerability → sublimation] *Response*: "世事变迁，不过又添诗料罢了。" (PC=0.71, declined but recovered DM)

Turn 20 (Recovery). PC returns to 0.83, within 5 turns as measured by protocol.

F Trigger Diagnostics Details

On a **diagnostic set** of 200 labeled interactions (100 critical, 100 non-critical) designed to cover the taxonomy in Section 3.2.1, our rule-based trigger achieves:

- **Precision**: 96.3% (when triggered, 96.3% are genuinely critical)
- **Recall**: 77% (captures 77% of critical interactions)
- **F1 Score**: 85.6%
- **Accuracy**: 87% with trigger rate 40%

Per-Category Analysis. Based on 200 labeled samples:

- **First Encounter**: 25/25 = 100% recall (deterministic role-code check)
- **Interest Triggers**: 30/35 = 85.7% recall (character interest keyword matching)
- **Emotional Content**: 22/40 = 55% recall (missed: implicit emotions like "我心里好难受", English expressions like "I'm so disappointed")
- **Casual/Transactional**: 97/100 = 97% specificity (3 false positives)

Learnable Trigger Experiment. Using our 200 labeled samples, we trained a lightweight BERT-base classifier:

- **Learned Trigger**: Precision 94%, Recall 87%, F1 90.2%
- **Trade-off curves**: At precision-matched threshold (98

We use rule-based triggers in main experiments for reproducibility.

F.1 Evaluation Validation Details

Multi-Evaluator Analysis. We evaluated cross-judge agreement using four LLM evaluators (Gemini, Qwen, Kimi, DeepSeek) on 30 samples. The pairwise Pearson correlations are:

- Gemini–Qwen: $r = 0.42$; Gemini–DeepSeek: $r = 0.52$
- Qwen–DeepSeek: $r = 0.12$; Kimi–DeepSeek: $r = 0.08$
- Average cross-judge correlation: $r = 0.15$

The relatively low inter-judge agreement reflects known variability in LLM-as-Judge scoring preferences—different models exhibit distinct rating biases (e.g., Kimi tends toward uniform mid-range scores). Importantly, despite absolute score differences, all evaluators consistently rank PersonaForge above baselines, supporting the robustness of our relative comparisons. We mitigate this limitation through human expert validation ($r = 0.82$) as the primary reliability metric.

Length Control Analysis. (1) Truncate all responses to 100 characters: correlation $\rho = 0.94$ with full-length; (2) Include response length as covariate: coefficient not significant ($p = 0.34$).

F.2 Defense Mechanism Validation Protocol

Two psychology graduate students independently annotated 100 "stressor-triggered" turns:

- LLM-as-Judge achieves $\kappa = 0.73$ with human experts on stressor detection
- $\kappa = 0.69$ on DM type classification

Blind Annotation Protocol. 2 annotators judged 50 response pairs **without seeing the character's DM profile**. Agreement with profile-informed judgments: $\kappa = 0.68$.

G State Update Sensitivity

- **Random stressor detection**: Drift rate 8.2% → 18.6%
- **BERT-base sentiment**: Drift rate 8.2% → 12.4%

- **No self-consistency voting**: Drift rate 8.2% → 10.8%

H Defense Mechanism Detailed Analysis

DM Ablation. Human raters (N=3) evaluated 60 response pairs: With DM 4.35 ± 0.42, Without DM 3.12 ± 0.68 ($p < 0.001$).

DM vs. Style Disentanglement. Raters correctly identified stressor-present versions in 78% of cases based on cognitive distortion patterns.

Mechanism Pair	Baseline	Ours
Ration. ↔ Intellect.	23%	8%
Sublim. ↔ Repress.	18%	5%
Displace. ↔ Project.	15%	4%
Activation Precision	61%	92%

Table 6: Defense mechanism confusion matrix and activation precision.

Confusion Matrix.

I Cross-Partner Evaluation Details

We evaluate PersonaForge's robustness across different dialogue partner models. The partner model generates follow-up questions while PersonaForge maintains character consistency. Table **??** shows results over 10-turn dialogues.

Partner	Avg PC	Drift
Gemini 2.5	**1.00**	**0.0%**
Kimi	1.00	0.0%
DeepSeek	0.98	-5.0%
Qwen-Plus	0.93	+5.0%

Table 7: Cross-partner evaluation. PersonaForge maintains high PC (0.93–1.00) across all partner models, demonstrating robustness to different interaction styles.

J Failure Case Mitigations

K Cost-Performance Analysis

Token Budget Fairness Controls.

- **Extended-Persona**: Structured-CoT with 850-token persona: PC 0.76 (vs. 0.86)
- **Memory-Augmented**: RAG-Persona with 5 examples: PC 0.74, drift 28.2%

Mitigation	PC	Drift
Baseline (no mitigation)	0.89	2.5%
+ Context-aware stressor	0.88	0.0%
+ Register-adaptive style	0.85	0.0%
+ Relationship priors	0.85	2.5%
All combined	**0.84**	**0.0%**

Table 8: Failure case mitigations impact. Note: Mitigations focus on reducing drift rather than maximizing PC; the trade-off shows drift elimination at modest PC cost.

Method	Tokens	PC	PC/kT
S-CoT	255	0.66	2.58
Ours (Sel.)	289	0.97	3.35
Ours (Always)	439	0.98	2.23

Table 9: Cost-performance comparison. PC/kT=PC per 1000 tokens.

L Additional Results

L.1 Open-Source Generator Validation

Generator	Ours	S-CoT	Δ
Gemini 2.5	0.86	0.72	+0.14
Qwen-Plus	0.80	0.64	+0.16
Kimi	0.73	0.53	+0.20
DeepSeek	0.83	0.45	+0.38

Table 10: Cross-generator PC validation.

M Cross-Domain Evaluation Details

N RoleBench Evaluation Details

Setup. 20 RoleBench characters, 20 turns each. Evaluation: Pairwise LLM-as-Judge.

O Failure Case Studies

Case 1: Stressor Misdetection. Cao Cao responding to subtle sarcasm. LLM misclassified indirect criticism as neutral, failing to activate Projection. *Root cause*: Subtle literary devices (irony) are under-detected.

Case 2: Style Constraint Conflict. Wang Xifeng in a formal negotiation. Her casual speaking style clashed with the scenario's formal register. *Root cause*: Style matrix lacks register switching.

Domain	Ours	S-CoT
Chinese (n=37)	0.86	0.72
Fantasy (Ice&Fire, n=16)	0.82	0.63
Whimsical (Alice, n=29)	0.84	0.65

Table 11: Cross-domain PC comparison.

Method	Win-Rate	Drift	Avg PC
RoleLLM-style	50.0%	24.8%	0.69
Structured-CoT	53.5%	20.4%	0.72
Ours	**73.2%**	**8.4%**	**0.81**

Table 12: RoleBench evaluation results (20 characters, 20 turns each).

Case 3: Relationship Cold-Start. New interlocutor with ambiguous role. Model defaults to neutral, missing character-appropriate warmth/suspicion. *Root cause*: Dynamic state initialization is simplistic.

O.1 Evaluation Bias Controls

We mitigate common LLM-as-Judge pitfalls:

1. **Closed-loop bias**: Human expert correlation ($r = 0.82$) and cross-generator consistency validation.
2. **Length bias**: $\rho = 0.94$ correlation with truncation; length is not significant ($p = 0.34$).
3. **Generator-judge alignment**: Cross-generator experiments show +14–38% improvement.
4. **Adversarial robustness**: 98% rejection of keyword-stuffed responses.
5. **State extraction**: BERT-base sentiment increases drift to 12.4%, showing graceful degradation.

O.2 Generalization Details

Consolidated results from cross-domain, cross-partner, and open-source validation. See Sections **??**, O, and **??**.

O.3 Broader Applications

- **Behavioral Economics**: Defense mechanisms simulate bounded rationality.
- **Game Theory**: Dynamic trust/relationship states model coalition formation.
- **Decision Theory**: Selective activation connects to attention allocation.

O.4 PC Threshold Sensitivity

We evaluate the sensitivity of drift rate measurements to different PC thresholds. Samples scoring below the threshold are counted as "drifted." Table **??** shows that PersonaForge achieves substantially lower drift rates across all thresholds, with particularly strong performance at the strictest threshold (0.5), where it achieves 0% drift compared to 1.25% for S-CoT.

Threshold	Ours↓	S-CoT	Δ
0.5	0.0%	1.25%	-100%
0.6	1.25%	7.5%	-83%
0.7	1.25%	7.5%	-83%

Table 13: Drift rate at different PC thresholds. PersonaForge (Ours) shows significantly lower drift across all thresholds, demonstrating robust personality consistency.

P Cognitive-Economic Optimization

Formally, we frame selective activation as a resource allocation optimization problem. Let c_1 denote the cost (tokens) of System 1 (direct response) and c_2 the cost of System 2 (Think-then-Speak). Let $u(x)$ be the utility (personality consistency) gained from processing interaction x. Our mechanism solves:

$$\max_{\pi} \sum_{x} \pi(x) \cdot u_2(x) + (1 - \pi(x)) \cdot u_1(x) - \lambda \cdot c(\pi) \tag{1}$$

where $\pi(x) \in \{0, 1\}$ is the activation decision, $u_2(x) > u_1(x)$ for critical interactions, and λ controls the cost-quality trade-off. Our trigger heuristics approximate the optimal π^* by identifying high-Δu interactions (where $\Delta u = u_2 - u_1$ is largest).

Q Failure Mode Comparison: Why Each Layer Matters

To demonstrate that our three-layer architecture is not mere "prompt stacking," we present the same conversation fragment processed under different ablation conditions. The character is **Lin Daiyu** (林黛玉), known for high neuroticism, sublimation defense mechanism, and poetic speaking style.

Scenario. Interlocutor criticizes her poetry: "你的诗虽然辞藻华丽，但缺乏深度。" ("Your poetry, though ornate, lacks depth.")

Full System Response. *Inner Monologue*: [Hurt → sublimation: redirect to creative process]
Output: "深浅由人评说，我不过借花草寄情罢了..." (PC=0.89, SA=0.85, DM=Sublimation correctly activated)

Failure Mode 1: Without Big Five. Output loses neurotic sensitivity, becomes generic: "谢谢指点，我会改进的。"
Collapse type: **Trait drift**—response could belong to any agreeable character. (PC=0.52)

Failure Mode 2: Without Defense Mechanism. Output shows hurt but no psychological coping: "你说的对，我确实不够好..."
Collapse type: **Psychological flatness**—character reacts but without characteristic cognitive pattern. (DM=None, perceived as out-of-character)

Failure Mode 3: Without Speaking Style. Output maintains personality but loses voice: "我只是用诗歌来表达内心感受而已。"
Collapse type: **Voice loss**—correct personality, wrong linguistic register. (SA=0.41)

Failure Mode 4: Without Dynamic State. Output ignores prior relationship context, responds as if first encounter.
Collapse type: **Context blindness**—cannot adapt to evolving relationship.

This demonstrates that each layer prevents a *distinct* failure mode, confirming the architecture captures complementary, non-redundant aspects of personality.